Inspection, traceability and quality control

USBGE-QA-001 | Document revision 04 | 2026-10-16

Design baseline USBGE-A-WT-20261016-R04 | A (native schematic title revision A0)

Quality and standards framework

Use IPC-6012 with IPC-A-600 for rigid PCB performance and incoming board acceptability; use J-STD-001 with IPC-A-610 for soldered assembly process and finished assembly acceptability. Quality and customer shall select the applicable revision/class in the purchase specification. This package neither assigns a default class nor claims certification.

Use ANSI/ESD S20.20 or IEC 61340-5-1 as the agreed ESD control framework; control moisture handling to applicable component supplier instructions and the agreed J-STD-033 process. Specific normative clauses require licensed copies and contractual applicability; this pack does not reproduce standard acceptance tables.

Inspection is 100% for the first article and the proposed initial build screen. Lot sampling, AQL and destructive microsection sampling are not defined by design evidence; quality shall approve a risk-based sampling plan before reducing coverage.

Incoming inspection work instruction

Record supplier, PO, part/revision, board lot/date code, quantity and certificates. Verify artwork ID, thickness, stack construction, finish and impedance report match the agreed fabrication order. Check packaging, contamination, conductor damage, exposed laminate, mask registration, hole damage and delamination under magnification. Measure the 70 x 34 mm contour and selected critical hole locations using calibrated equipment against approved tolerances.

Review electrical test certificate and coupon results, including test method and actual measured impedance. Without an agreed thickness/finish/tolerance/class or traceable evidence, quarantine the lot. Record observations and photographs without inventing a pass.

PCBA inspection and electrical screening

Reconcile all 48 references against master BOM and material lots. Check orientation with numbered pad details, component seating, pin/lead damage, solder bridges, insufficient wetting, tombstones, balls/debris, exposed-pad quality, connector anchor joints and barrel fill. Check residual flux/cleanliness against the qualified process.

Inspection records shall identify magnification, AOI program/revision, X-ray image paths and reviewer. U2/U3 hidden joints require a qualified inspection method; do not accept a device solely because an exterior fillet is visible. Follow PT-04 before applying power and PT-05 for every qualified power-up.

Serialisation and traceability

Proposed physical identifier format: USBGE-A-YYWW-NNNNN, issued sequentially by the controlled register without reuse. Quality shall approve label size/location and durability clear of high-speed contacts, exposed pads and heat paths. A printed serial label is distinct from a programmed USB serial/MAC.

For each serial retain PCB fabrication lot, assembly lot/traveler, manufacturer/date/lot codes for controlled parts, BOM revision, stencil/profile revisions, operator, inspection/test results and attachments, station/software IDs, calibration status, NCR/rework links and final disposition. Retention duration and access rights are organisation decisions; assign them before production release.

Nonconformance and rework

NCR sequence: identify and segregate affected material; record serial/lot, symptom, requirement, evidence and detected stage; assess scope to other lots; obtain engineering/quality disposition; implement only authorised rework; retest impacted functions plus regression; record verification and closure. Permissible dispositions are rework, return, scrap or specifically authorised concession. No operator use-as-is decision.

Rework instructions must specify component order code, method, profile/temperature limits, inspection and retest. Record each cycle and board damage. Concessions must name affected quantities/serials and expiry; they do not silently revise the master design.

Engineering change control

Open ECR with reason, evidence and affected baseline. Engineering evaluates electrical, mechanical, SI/EMC, test, tooling, supplier, stock and documentation impacts. ECO approval identifies precise native files, revised documents, validation requirements, effectivity and stock disposition. Implement on a controlled new baseline; regenerate outputs; rerun validation; obtain genuine release approval before use.

Approved C5 parameters/comment were updated in the authoritative source with before/after evidence. All unrelated source content is preserved. Fresh Rev 04 Altium compile/ERC/DRC and native exports have not completed: startup produced no usable window or script-entry evidence. Rev 03 engine reports are historical evidence. Local source/stream checks are not ERC/DRC. No physical validation or formal release approval is claimed.

Referenced sources

ID / reference	Source and scope
SRC-01 / Microchip LAN7800 datasheet DS00001992J (2025)	https://ww1.microchip.com/downloads/aemDocuments/documents/UNG/ProductDocuments/DataSheets/LAN7800-Data-Sheet-DS00001992.pdf Table 3-1 pin polarity; section 16.2 operating rail limits; section 11 OTP context. Accessed 2026-10-16.
SRC-02 / Texas Instruments TLV62568 datasheet Rev B	https://www.ti.com/lit/ds/symlink/tlv62568.pdf Feedback model and DBV package context. Accessed 2026-10-16. Exact assembly qualification is not inferred.
SRC-03 / IPC document revision table	https://www.ipc.org/ipc-document-revision-table Official standards catalog; select contractual revisions with quality, not automatically latest. Accessed 2026-10-16.
SRC-04 / IPC J revisions assembly standards announcement	https://www.ipc.org/news-release/ipc-releases-j-revisions-two-leading-standards-electronics-assembly Distinguishes soldering process requirements J-STD-001 and post-assembly acceptance IPC-A-610. Accessed 2026-10-16.
SRC-05 / KiCad Altium binary reader implementation	https://raw.githubusercontent.com/KiCad/kicad-source-mirror/master/pcbnew/pcb_io/altium/altium_parser_pcb.cpp Field-layout cross-reference for bounded local pad/via/track/arc extraction. Not a native Altium validation engine. Accessed 2026-10-16.
CS-01 / TI TLV62568 SLVSD89B	https://www.ti.com/lit/ds/symlink/tlv62568.pdf Sections8.2.2.3-5: LC matrix and capacitor/current sizing. Local PDF/hash. Fitted 4.7 uH is outside cited 1/2.2 uH characterized table; no stability pass inferred.
CS-02 / Microchip LAN7800 DS00001992J	https://ww1.microchip.com/downloads/aemDocuments/documents/UNG/ProductDocuments/DataSheets/LAN7800-Data-Sheet-DS00001992.pdf Tables3-1/3-2 and section 4: pin 38 VDD33A supply. Local PDF/hash.
CS-03 / USB-IF Link Layer Test Rev 1.1	https://www.usb.org/sites/default/files/USB%203%202%20Link%20Layer%20Test%20Specification%202024%2003%2003.pdf Test6.1 receiver polarity correction rationale; no board compliance claim.
CS-04 / Murata 22 uF / 0603 typical curves, distributor-hosted manufacturer PDF	https://www.farnell.com/datasheets/4459590.pdf Murata typical curves page 3: approximately 8-10 uF each at operating bias. Used only for a conditional two-bulk-cap bank scenario; not guaranteed bounds or proof of actual C14 part.
CS-05 / Murata 47 uF / 0805 typical curves, distributor-hosted manufacturer PDF	https://www.farnell.com/datasheets/4459607.pdf Manufacturer identity/typical curves of considered47 uF / 0805 option. Not preferred with current C14: nominal bank69.5 uF exceeds cited TI guidance.
CS-06 / Murata GRM21BR61A476ME15 series official specification	https://search.murata.co.jp/Ceramy/image/img/A01X/G101/ENG/GRM21BR61A476ME15-01A.pdf Manufacturer identity/typical curves of considered47 uF / 0805 option. Not preferred with current C14: nominal bank69.5 uF exceeds cited TI guidance.
CS-07 / Altium pick/place centre definition	https://www.altium.com/documentation/knowledge-base/altium-designer/calculate-component-pick-and-place-center-of-a-footprint Midpoint of outermost pad-centre positions, not reference origin.

ID / reference	Source and scope
CS-08 / Altium native API	https://www.altium.com/documentation/altium-dxp-developer/workspace-manager-api Native compile/outputer/compiled pin-net APIs. Exact executable scripts retained.
CS-09 / Altium PCB interfaces	https://www.altium.com/documentation/altium-dxp-developer/pcb-api-system-interfaces-reference Board output/Gerber/DRC option interfaces; generator Boolean alone is not output completeness.
CS-10 / User engineering update / saved files	Hardware/Altium and this chat User states pin 38 fixed in schematic/PCB and DRC zero errors. Saved source hashes, native log Added Pin To Net +3V3 U3-38 and 14:39:45 DRC retained; follow-up report14:55:39 binds fresh copied current baseline.